

Yujie Tang

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SUMMARY

A self-disciplined and curious graduate student with a strong foundation in **robotics and control**. Passionate about programming and robotic path planning, with particular interest in safe and autonomous control algorithms. Experienced in developing C, C++, Python, and MATLAB programs, and skilled in simulation and debugging across diverse platforms including Linux, Arduino, and STM32

EDUCATION

2024 - present Master's Degree at **Harbin Insititute of Technology**, School of Robotics and Advanced Manufacture, in Control Engineering (GPA: 3.83/4.4, Top 5.5%)
2020 - 2024 Bachelor's Degree at **Beihang University**, School of Mechanical Engineering and Automation, in Robotics Engineering (GPA: 3.68/4.0, Top 31.8%)

PUBLICATIONS

Yujie Tang et al. (2025). "Flexible and safe navigation of autonomous underwater vehicles with input-dynamics move blocking". In: *Ocean Engineering* 339, p. 122182. URL: <https://www.sciencedirect.com/science/article/pii/S0029801825018669>

- **Key contributions:** Developed an algorithm to improve MPC solution efficiency for AUVs in disturbed environments. Added dynamic parameters to CBF for earlier detection and avoidance of moving obstacles. Integrated a disturbance observer to construct a complete control system with code released on [Github](#)
- **Personal Rules:** Designed the algorithms and experimental setup; Conducted experiments; Wrote paper

INTERNSHIP & PROJECT

Meituan—Autonomous Vehicle Decision and Planning (Intern) Nov 2025 - Feb 2026

- **Data Processing:** Supported data processing tasks for autonomous vehicle decision-making strategies; used SQL to perform multi-table joins, JSON field parsing, and aggregation metric construction
- **Data Visualization:** Built interactive visualization pages using Mercury
- **Team Collaboration:** Recorded issues and detailed solution steps during work; organized documentation

Robocup Soccer—Humanoid League

- Participated in robot motion control debugging, documentation, and code management; Studied biped robot walking stability theories and project structures; derived camera coordinate transformation formulas; Implemented PID control based on odometry for walking commands

MAIN HONORS & AWARDS

- **(National)** National Scholarship Oct 2025
- **(Harbin Institute of Technology)** Graduate Academic Excellence Scholarship, Special Class Oct 2025
- **(National)** RoboCup China Open, Second Prize Nov 2022
- **(Provincial)** Asia-Pacific Robot Cup International Invitational, First Prize May 2023
- **(Provincial)** Robot Competition for College Student in Five Provinces of North China, First Prize Nov 2022
- **(Beihang University)** Outstanding Graduate Jun 2024

In total, received 17 other scholarships and competition awards

SKILLS

- **Tools:** Git, Latex, Markdown; Typora, Notion. Quick to adopt new tools through self-study
- **Language:** CET-6: 567
- **Core courses:** Linear Algebra (96), Automatic Control (93/96), Fundamentals of Robotics (97), Interchangeability and Tolerance Design (97), Theory of Machines (93), Engineering Graphics (92)
- **Robot Dynamics Simulation (MATLAB):** Modeled single- and double-joint robots; implemented feedforward control, inverse dynamics, and hybrid force–position controllers.
- **Game Development (C++):** Built a structured piano-tile game using EasyX graphics library and data structures.